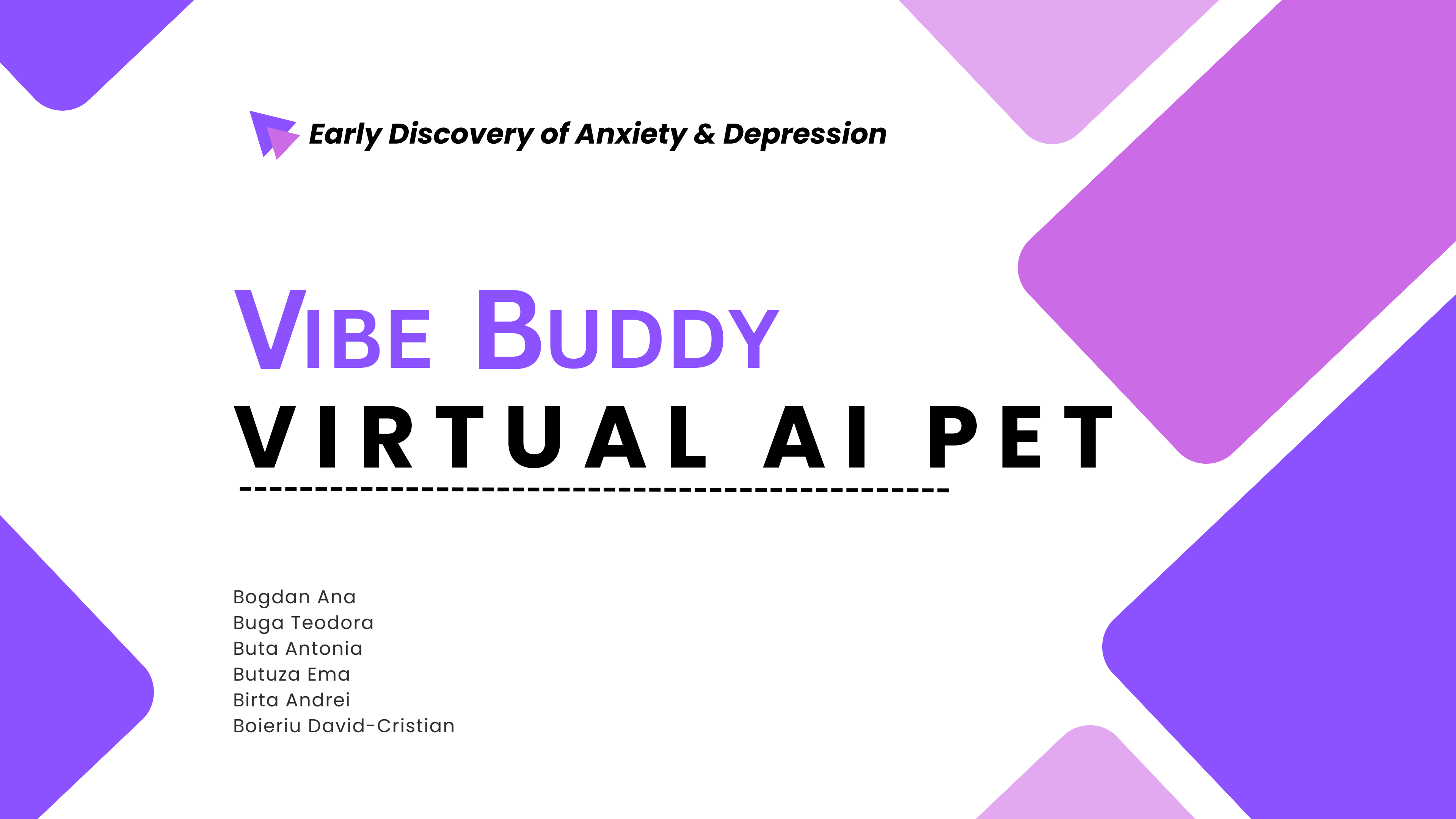




 ***Early Discovery of Anxiety & Depression***

VIBE BUDDY

VIRTUAL AI PET

Bogdan Ana
Buga Teodora
Buta Antonia
Butuza Ema
Birta Andrei
Boieriu David-Cristian



1. BUSINESS PART

MARKET OVERVIEW

- **Market** : Digital mental health and wellness apps
- **Global mental health app market**: ~\$6–7B (2024)
- **Expected growth**: 15–20% CAGR
- **Core growth drivers**: Gen Z, mobile-first habits, mental health awareness
- **Targetable users** (teens/young-adults): ~500M

Segmentation	Description
Geographic	Urban/Suburban
Demographic	13–22 years old
Psychographic	Emotionally aware, digital-native
Behavioral	Daily digital users

INDUSTRY & COMPETITIVNESS ANALYSIS

Industry characteristics:

- Emotionally driven decisions
- High trust and privacy concerns
- Retention > acquisition
- AI + UX differentiation is critical

Our approach to build competitive advantage:

- Emotional attachment
- Empathic AI
- Non-clinical, youthful design

Competitors Comparison				
Feature	Our app	Woebot	Calm	Headspace
AI Chat	 Yes ▾	 Yes ▾	 No ▾	 No ▾
Virtual Companion	 Yes ▾	 No ▾	 No ▾	 No ▾
Youth-focused Design	 Yes ▾	 ▾	 No ▾	 No ▾
Breathing Exercises	 Yes ▾	 ▾	 Yes ▾	 Yes ▾
Daily Motivation	 Yes ▾	 ▾	 ▾	 ▾
Emotional Bond	 Yes ▾	 No ▾	 No ▾	 No ▾

VALUE PROPOSITION

- **Core Statement:** “Your emotions deserve a friend”
- B2C product
- **Value proposition #1:**
→ target: teen/student user
- **Value proposition #2:**
→ target: parents/guardians, psychology professionals, schools

Customer need	Our response
Fear of judgement	Private AI pet
Low motivation	Streaks and rewards
Emotional confusion	AI insights
Inertia (using nothing)	Fun entry point

BUSINESS MODEL

Monetization model:

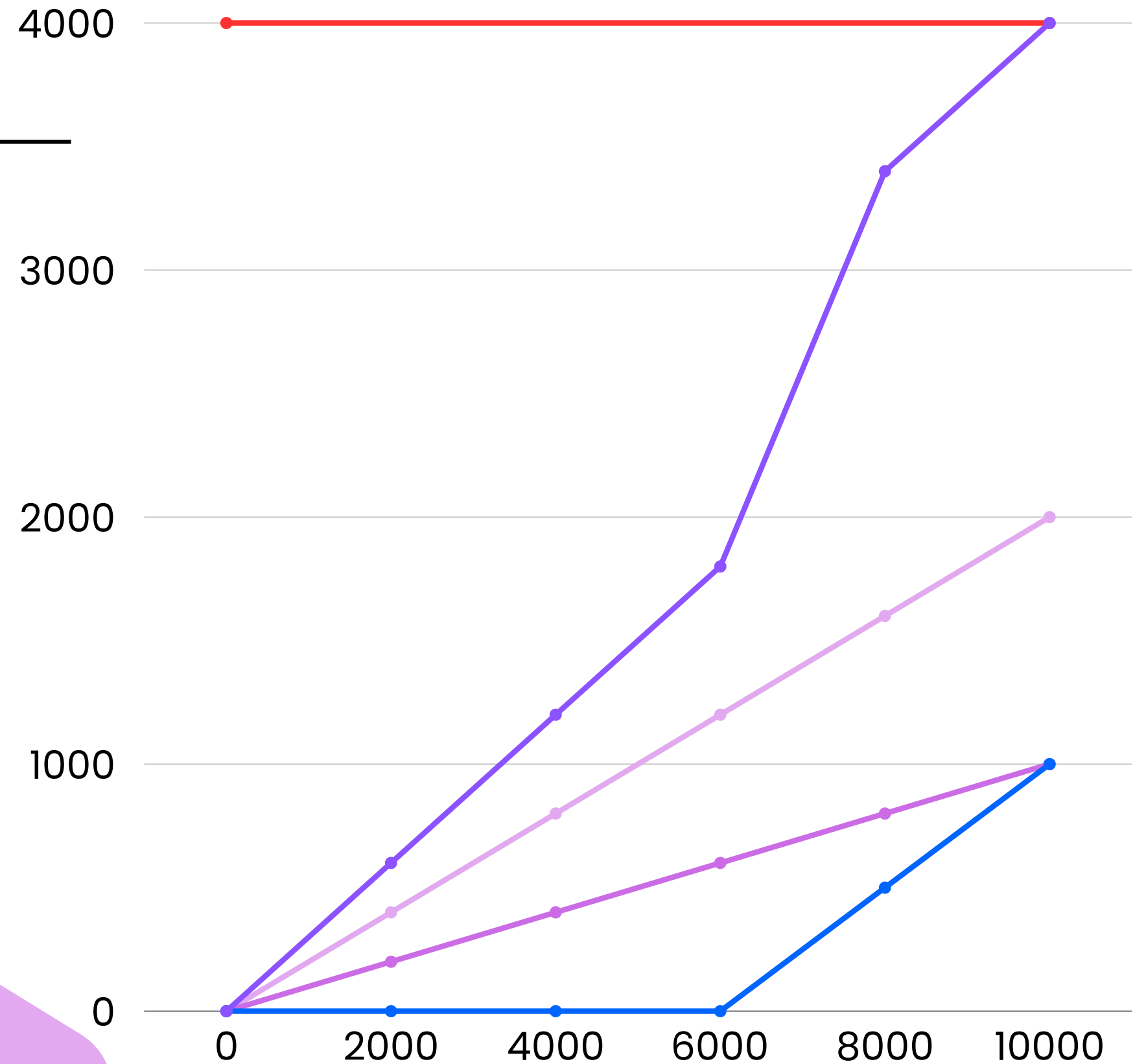
- 100% free access for teens and students
- revenue streams: professional referrals and partnerships
- schools or NGOs fund access for students

Costs/Month:

- AI/Cloud usage: 1800\$
- Devs and maintenance(small team): 1500\$
- Ops and misc.: 700\$

Break even:

- Ads (0.20\$ active user/month)
- Professional partnerships revenue (100\$ per professional/month - 1 professional/1000 users)
- School/ONG funds (once app grows)





2. TECH PART

FRONTEND

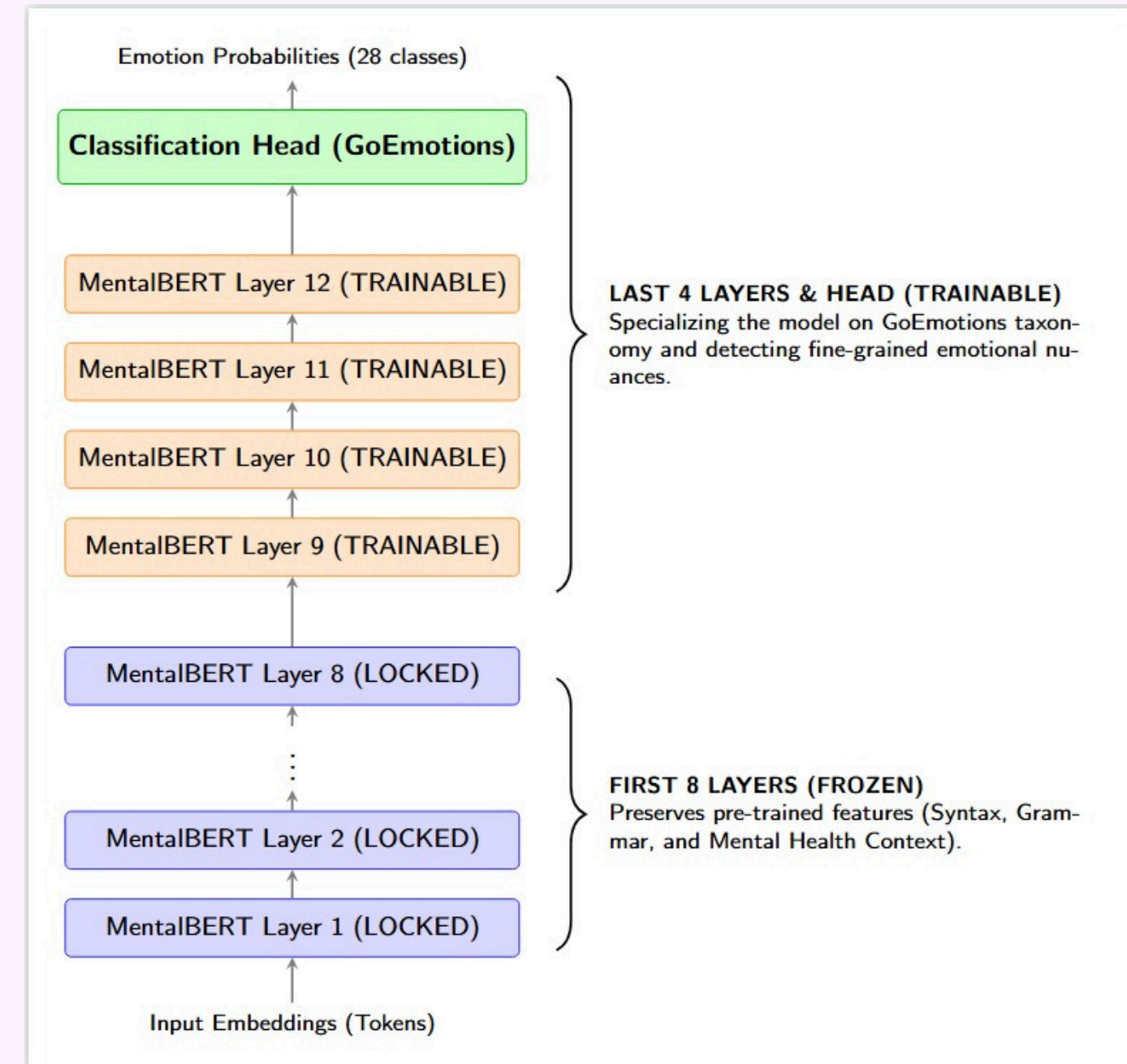
- **Framework:** React 19 with TypeScript
- **Build Tool:** Vite
- **Styling:** Tailwind CSS with custom design system
- **UI Components:** shadcn/ui (Radix UI primitives)
- **Animations:** Framer Motion
- **Icons:** Lucide React
- **Charts:** Recharts for progress visualization
- **State Management:** React Hooks (useState, useEffect)

BACKEND

- **Framework:** FastAPI (Python)
- **LLM:** Phi-3 via Ollama for empathetic chatbot responses
- **Emotion Detection:** Custom-trained MentalBERT model for fine-grained emotion detection (28 emotions)
- **Database:** SQLite with SQLAlchemy for persistent conversation storage
- **Memory:** Conversation and emotional state tracking with persistent history

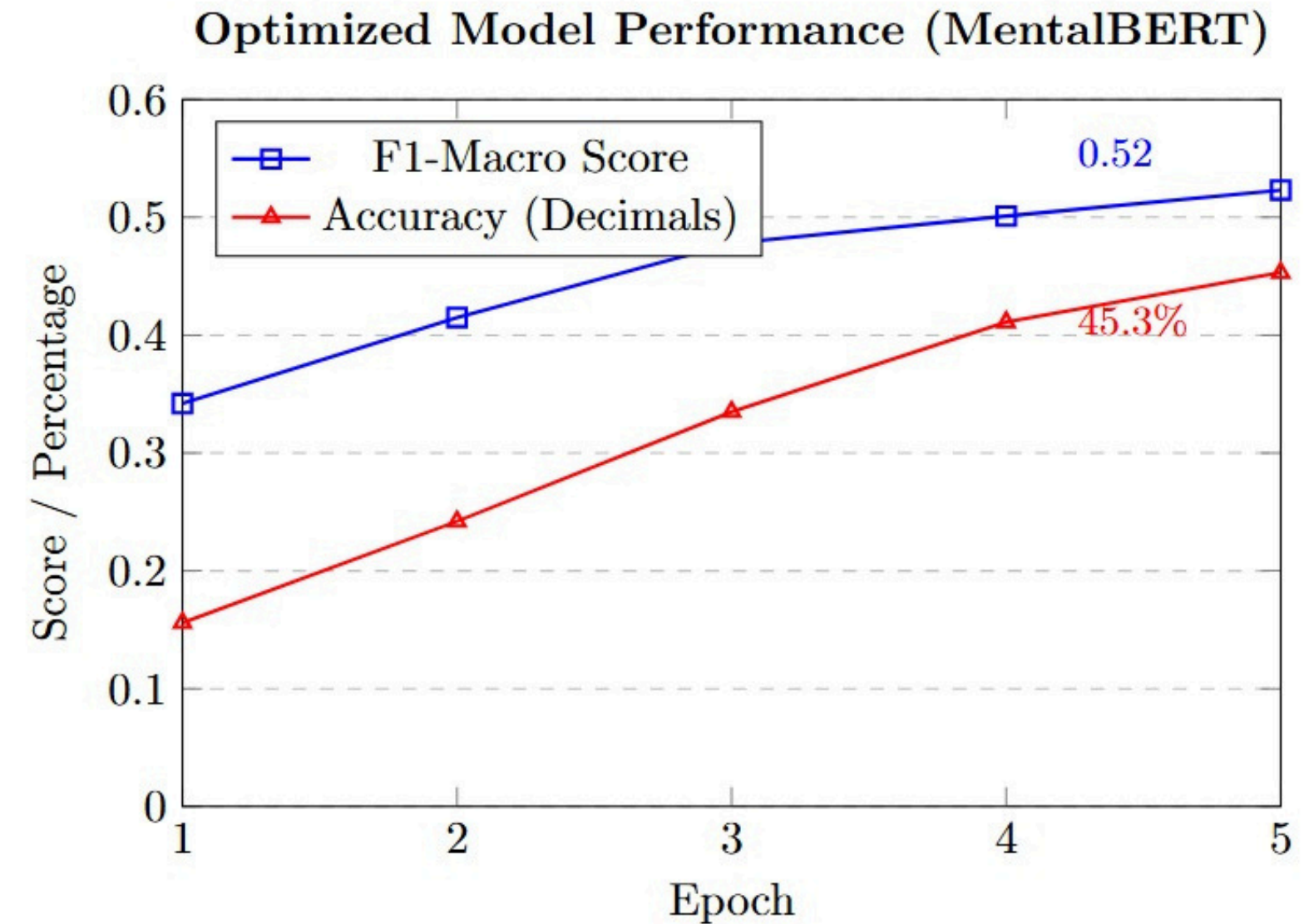
AI Integration

- **Semantic dilemma:**
 - Limitations of trained models on encyclopedic corpora
 - Importance of Domain adaptive pre-training
 - Negative emotions sensitivity
- **Transfer Learning strategy:**
 - Catastrophic Forgetting prevention
 - Training time reduction : > 60%
 - High level emotional features



RESULTS

- Wighted BCE loss: compensation for dataset imbalance
- Metrics interpretation : 28-class taxonomy, an F1 score of 0.37 => discriminative capability superior to chance (baseline)
- Qualitative success : Correct correlation of complex emotions



The background features two large, light purple, rounded triangular shapes in the corners. One is in the bottom-left corner, pointing towards the center, and the other is in the top-right corner, also pointing towards the center.

CONCLUSIONS

 **THANK YOU!**
LET'S SHOW THE DEMO!

